

MUKUL MUNDLE

Principal Data Scientist

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EXECUTIVE SUMMARY

A results-driven Principal Data Scientist with 6+ years of experience architecting and deploying high-impact AI/ML, Generative AI, Time Series Forecasting, Deep Learning, Data Engineering, and Cloud (AWS, Azure, GCP) solutions across the Automotive, Pharmaceutical, and BFSI sectors. Managed the end-to-end project lifecycle & skilled in sophisticated RAG architecture, LLM fine-tuning, and multi-agent AI systems deployment to solve complex business problems. I have a proven track record of delivering significant value, from enhancing customer support with multimodal AI agents at Toyota to optimizing multi-million-dollar inventories at Sun Pharma & process Optimization at Dr. Reddy's and Jubilant. I am passionate about building scalable, enterprise-grade AI systems and ensuring their ethical deployment by adhering to Responsible AI standards.

KEY AI/ML SKILLS & TECHNOLOGIES

- Generative AI:** RAG, Fine-tuning, LLMs, LangChain, Llama-Index, Vector DBs, Prompt Engineering, Diffusion
- Deep Learning:** NLP, Computer Vision, TensorFlow, PyTorch, ONNX, ISO/IEC 42001 Principles, TimeGPT, Transformers, TimesFM
- Cloud & BI Tools:** AWS (SageMaker, Lambda, EC2, S3, Redshift), Azure (AI, Databricks), GCP (Vertex AI, Big Query), PowerBI, Tableau, QlikView
- AI Agents:** AutoGen, LangGraph, CrewAI, Multi-Agent Systems, Tool Integration, Multimodal Models
- Machine Learning:** XGBoost, LightGBM, Random Forest, SVM, Prophet, ARIMA, Gurobi, SciPy, Scikit-learn, Clustering, Anomaly Detection
- Data Engineering & MLOps:** Python, R, Databricks, Spark, Airflow, Docker, GitHub Actions (CI/CD), FastAPI, Weights & Biases (WandB), SQL, PostgreSQL (PostGIS), ETL, JIRA

ACADEMIC QUALIFICATIONS

Indian Institute of Management (IIM) Calcutta - Kolkata	08/2022-03/2024
MBA(PGDBA): Business Analytics	
GPA: 8.56/10	
Indian Institute of Technology Kanpur - Kanpur	07/2013-05/2018
Integrated B. Tech – M. Tech: Chemical Engineering	
GPA: 8/10	
Shivaji Science College - Nagpur	2013
Grade 12th: HSC Board Maharashtra	
Percentage: 87.17 %	
Wainganga Vidyalaya	2011
Grade 10th: SSC Board Maharashtra	
Percentage: 96.73 %	

WORK EXPERIENCE (6 YEARS)

Ascentt (Client: Toyota Motors North America) - Pune

07/2024 – Current

Principal Data Scientist

- Led a team of 3 Data Scientists, managed project timelines (JIRA), and provided technical mentorship.

Advanced RAG for Customer Support

- Architected an advanced, multimodal RAG pipeline integrating text (manuals), images (diagrams), and structured data (vehicle specs) from disparate sources.
- Implemented a hybrid retrieval strategy using BM25 for keyword search and OpenAI/CLIP embeddings for semantic and image-based search with re-ranking
- Exposed the RAG system via FastAPI as scalable REST endpoints on Azure, reducing query resolution time by 40% and ensuring factual consistency, evaluated using the RAGAS framework.

Autonomous AI Agent Workflows

- Designed and deployed multi-agent systems using AutoGen and CrewAI to automate complex workflows like competitor analysis, customer issue resolution.
- Developed specialized agents with distinct roles (e.g., Planner, Information Retriever, Data Analyst) that collaborate to perform tasks, reducing manual effort in market research by 70%.
- Utilized LangGraph to build a stateful, agentic framework with integrated tools (vector search, search API), improving first-contact resolution rates by 40%.

Credit Risk Modelling (Toyota Finance)

- Developed an XGBoost model to predict auto loan defaults, achieving 89% AUC-ROC and reducing portfolio default risk by over 12%.
- Engineered a real-time early warning system to identify delinquency risk, enabling targeted outreach that reduced 30+ DPD accounts by 18%.

Geospatial Warehouse Optimization

- Analyzed inventory distribution and forklift movement patterns using PostGIS/PostgreSQL and developed an optimized routing algorithm, reducing travel distance and streamlining pick-and-pack processes.

Smart Forecasting and Inventory Optimization

- Led development of sales & inventory forecasting models (TimeGPT, NBEATS, Prophet) deployed on AWS Sagemaker with GitHub Actions.
- Implemented inventory optimization using Gurobi and orchestrated scalable data pipelines on Databricks with Spark, using Airflow for scheduling & WandB for model versioning.

Computer Vision for Defect Detection

- Developed and deployed a YOLO-based computer vision solution to identify and classify defects in Toyota cars, achieving 95% detection rates and reducing manual inspection time by 60%.

Sun Pharmaceuticals Industries Ltd. - Mumbai

10/2023 - 06/2024

Senior Manager MD's Office Data Scientist and Analytics

Domain-Specific RAG for Regulatory Search

- Built a RAG-powered knowledge engine with OpenAI text-embedding, Pinecone, and LangChain for semantic search across thousands of regulatory SOPs and clinical trial protocols.
- Evaluated and improved the system for factual accuracy and completeness using metrics like BLEU and ROUGE, enabling faster access to compliance information.

LLM Fine-Tuning for SOP Automation

- Fine-tuned LLaMA and GPT models using QLoRA on a vast corpus of internal pharmaceutical documents to automate SOP summarization, compliance checks.
- Captured pharma nuances, cutting manual review & documentation effort by 40%.

NLP for Clinical Trial Matching & Social Media Analytics

- Developed a GenAI pipeline using BioNER and LangChain to extract structured data (eligibility, dosage) from unstructured PDFs.
- Utilized BERT & LSTM for text classification to determine if a patient meets clinical trial eligibility criteria, streamlining the patient matching process.
- Utilized BERTopic for topical modeling, executed sentiment analysis models on social media data, providing valuable market

Inventory & Dynamic Price Optimization

- Reduced stockouts and excess inventory by \$30M through ML-driven inventory optimization (ARIMA, Prophet, XGBoost), cutting provisions by 10%.
- Developed SQL-based anomaly detection system & PowerBI based control towers to identify irregularities in the supply, demand, BOM, recipe master data
- Implemented a dynamic pricing engine using price elasticity modeling and linear programming (SciPy), delivering a 5-10% margin uplift.

Jubilant Life Sciences - Noida

08/2021 - 08/2022

Deputy Manager Digital Excellence and Data Science

AI for Manufacturing Excellence (\$1M Savings)

- Led the digital transformation strategy, creating a manufacturing ecosystem driven by AI and Industry 4.0 for India and North America.
- Deployed predictive maintenance models (SVM, ANN, Random Forest) that reduced monthly maintenance expenditure by 50% and improved MTBF by 20%.
- Enhanced product robustness by implementing DMAIC methodology and ML models (Random Forest, GBM), reducing batch failures by 30%

Dr. Reddy's Laboratories - Hyderabad

07/2018 - 07/2021

Business Analyst Data Science

Yield Improvement & Process Optimization

- Designed and led the ML pipeline for Yield Improvement, increasing the yield of the top 20 SKUs to 98% and delivering annual savings of ₹5.6 Cr.
- Utilized explainable AI techniques (SHAP, LIME) and anomaly detection (K-means, DBSCAN, Isolation Forest) to monitor critical process parameters, reducing the cost of poor quality with a potential impact of ₹3.3Cr.

Digital Twins & Energy Forecasting

- Architected a multi-constrained schedule optimizer using Gurobi and a Genetic Algorithm for a product mix of 200+ SKUs, reducing logistics costs by 22%.
- Developed power consumption time series model (SARIMAX, ARIMA) for predicting bid quantity from IEX (0.3Cr INR)

CERTIFICATIONS

- AWS Certified Machine Learning – Specialty
- LLM Engineering: Master LLMs, GenAI and AI Agents
- DeepLearning.AI Specializations, Generative AI for Everyone
- Data Engineering with Pyspark, Airflow, Azure with Databricks and Data Factory

ACHIEVEMENTS

- **Data Science Competitions:** Winner of DESIS Europe Mobility Index time series forecasting by DE Shaw & Housing Price Prediction case by American Express
- **Research:** Successfully defended master's thesis on PCA of the Signaling Network
- **Corporate:** Honored with Dr. Reddy's Flagship Young Leaders Program, CEO award for AIML Modelling contributed to WEF Global Lighthouse recognition.